The Law of Graphic Equalisation — Passive Continuation Grounding:
Momentum, Gravity, and the Cost of Alteration

Matthew Parslow
Independent Researcher

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DRAFT — draft-2026-09-06.1 r12 (6 September 2026). This is Paper 01i, a formal companion to the Graphic Equalisation, physics-grounding, total-perspective, information-gravity, and Narrative Momentum papers. It formalises a research and derivation ledger, not a new mechanics or gravitational theory. Its sole positive internal result is class-relative: reuse of declared carried structure has zero *additional* alteration cost. The paper derives no physical momentum, four-momentum, metric, connection, Einstein equation, stress–energy source, equivalence principle, momentum-conservation law, geodesic equation, or quantum-gravity result. Flat transport and load-shaped free propagation are recovery targets for a concrete realization with stated equations, limits, and measurements.

Abstract

Passive continuation is defined here as admissible reuse of presently operative structure with no additional alteration cost in a declared grounding class. It is not no activity, no time, no interaction at every grain, or no support cost in another frame. This strict distinction permits a comparison with two established targets without identifying their physical roles: momentum is not a force, and general-relativistic free fall is not a locally felt gravitational force on an ideal test body. In a flat regime a realization must establish measured four-momentum transport, $dp^\mu/d\tau = 0$, from a state map, dynamics, symmetry, and isolation conditions. In a load-shaped regime it must establish $u^\nu \nabla_\nu u^\mu = 0$ from a Lorentzian metric and its connection, a proper-time observable, and matter coupling, while separately supplying a source law. Neither target follows from the inherited project vocabulary. The conditional reconciliation investigated here is therefore exact: flat inertial transport and load-shaped geodesic motion are two physical readings of one passive-continuation law only if one concrete realization recovers their equations, source/exchange accounting, universal-coupling limits, and stated observables. A four-way ledger records what is inherited, hypothesised, still owed as a physical bridge, and constrained by experiment.

1 Scope and terminology

The question is not whether momentum and gravity are one force. They are not. Physical momentum is a state/charge-like quantity; force is a specified rate of its transfer or change. In general relativity (GR), an ideal freely falling test body has zero proper acceleration, while a supported body records a nongravitational support force. Curvature and tidal effects remain physically measurable (Einstein, 1916; Geroch and Jang, 1975; Will, 2014).

The question is narrower. Can one implemented law have a flat reading in which carried physical momentum selects inertial transport and a load-shaped reading in which geometry selects free propagation? This paper fixes the conditions under which that question becomes a

physics claim. It supplies neither the implementation nor an inference that the conditions are already met.

The inherited corpus uses frame-indexed graph, admissibility, operative history, and alteration-cost vocabulary. Its physics companion offers a change-budget comparison; its Narrative Momentum companion distinguishes preservation from alteration; its information-gravity companion requires typed sources, geometry, clocks, and observables before a physical reconciliation claim. The Stage-1 research record and Stage-2 derivation ledger for this paper show that none provides the missing physical maps. Momentum and gravity are historically distinguishable readings of a presently inseparable passive continuation: a present in-frame record can measure a declared local relation, but it cannot by itself attribute why the continuation is carried. That attribution requires a history-bearing inference record with stated support, source, and comparison conditions. It does not make momentum, gravity, inertia, curvature, or force identical.

Definition 1.1 (Declared continuation situation)**.** *A frame-indexed continuation situation is*

$$\mathfrak{C}_t^W = (W, G_t, \Gamma_t, \mathcal{P}_t, S_t, \mathcal{A}, \kappa, \mathcal{R}_t, \mathcal{O}_t),$$

where W is the frame, G_t the represented graph, Γ_t its admissibility grammar, $\mathcal{P}_t$ its admissible continuations, S_t the presently operative carried structure when represented, $\mathcal{A}$ a declared alteration class, κ its ordered cost domain, $\mathcal{R}_t$ relevant support, and $\mathcal{O}_t$ its record relation. This is project-level typed vocabulary; it is not a phase space, action, metric, clock, or source tensor.

An alteration class must declare its support, equivalence relation on alterations, comparison currency κ, realised reference condition, resource budget, admissibility and measurement procedure. If any required component is unavailable, the corresponding class cost is undefined, not silently zero. The zero in the reuse convention belongs only to a fully instantiated class.

Definition 1.2 (History-bearing attribution record)**.** *For an interval $[t_0, t]$, a* history-bearing attribution record *is*

$$\mathcal{H}^W_{[t_0,t]} = \langle (\mathcal{O}_s, S_s, \mathcal{R}_s)_{t_0 \le s \le t},\ I^W,\ \Lambda^W \rangle,$$

where the retained sequence may be compressed but must preserve the stated changes, supports, and comparison relations relevant to the typed attribution map

$$I^W : ((\mathcal{O}_s, S_s, \mathcal{R}_s)_{t_0 \le s \le t}, \Lambda^W) \longrightarrow \mathfrak{A}^W_{\mathrm{phys}}.$$

The codomain $\mathfrak{A}^W_{\mathrm{phys}}$ is a declared set of typed physical attributions (for example, momentum transport, load-shaped continuation, curvature, or source claims), with their required observable and comparison models. Λ^W states the record's grain, losses, uncertainty, and domain of use. The record is neither the total past nor a view from nowhere. It is the minimum kind of record from which W can attempt to attribute a present passive continuation to carried momentum-like history, load-shaped continuation conditions, or both.

2 Inherited free-continuation and change-load claims

Where κ is additive, write

$$C_{\mathcal{A}}(a \mid S_t) = C_{0,\mathcal{A}}(a) + \Delta C_{\mathrm{alt},\mathcal{A}}(S_t, a).$$

The inherited distinction licenses the following definition and no stronger physical conclusion.

Definition 2.1 (Passive continuation)**.** *An admissible continuation $a \in \mathcal{P}_t$ is* passive within $\mathcal{A}$ *when it reuses the declared carried structure and*

$$\Delta C_{\mathrm{alt},\mathcal{A}}(S_t, a) = 0.$$

For an ordinal cost domain the valid statement is only the corresponding no-extra-alteration ranking. Passive continuation means no additional *alteration cost in $\mathcal{A}$; it does not mean zero total activity, zero total cost, zero elapsed or inferred time, zero environmental support, or zero interaction at every descriptive grain.*

Proposition 2.2 (Internal passive-reuse result)**.** *Given Definition 2.1, an admissible reuse continuation has zero additional alteration cost in its declared class.*

Proof. The conclusion is the defining cost assignment for reuse in that class. It is not a conservation theorem because no physical state, unit, dynamics, symmetry, or boundary current occurs in the premises. □

Proposition 2.3 (History required for physical attribution)**.** *A present passive-continuation relation does not determine a unique attribution to physical momentum or load-shaped gravity. Any such attribution requires a history-bearing record $\mathcal{H}^{W}_{[t_0,t]}$ and a declared inference map from that record to the relevant physical observables.*

Proof. Passive continuation classifies whether the present carried structure is reused; it contains neither a physical momentum map nor a source/geometry map. A present record can therefore be compatible with more than one physical realization. The record in Definition 1.2 supplies the retained changes, supports, and comparisons needed to test a proposed factorization, but still does not supply the required physical map. Thus history is necessary for attribution within this programme, though not sufficient without the declared inference map. □

Remark 2.4 (What in-frame inseparability does and does not say). An unspecified in-frame record supplies no physical interpretation. With a declared local accelerometer observable, a frame can determine proper acceleration and distinguish free from supported motion in its stated domain. That result does not differentiate momentum and gravity as explanatory factors: both can remain readings of the present passive continuation. Their attribution requires the history-bearing record of Definition 1.2, together with stated source, support, and comparison models. A local accelerometer cannot thereby infer curvature or source structure, or distinguish a homogeneous gravitational description from uniform acceleration without the specified extended, source, or cross-frame inference. The limitation is therefore local and operational: extended tidal measurements, support-force readings, source modelling, and cross-frame comparison contribute to the required historical inference. In particular, local inertial coordinates can remove connection coefficients at one event, not curvature across a region (Einstein, 1916).

3 Standard physical targets

Newtonian mechanics distinguishes momentum from force:

$$\mathbf{p} = m\mathbf{v}, \qquad \mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt}.$$

In an inertial frame, a free particle has constant momentum. For a closed system, total momentum conservation additionally requires an appropriate action or balance law, translation symmetry, and boundary-flux accounting; it does not follow from persistence language alone (Newton, 1687; Noether, 1918).

Given a Lorentzian metric, a proper-time functional, and the point-particle action $S_{\text{pp}} = -mc\int d\tau$, the GR free-test-body target is

$$u^{\nu}\nabla_{\nu}u^{\mu} = 0, \qquad u^{\mu} = \frac{dx^{\mu}}{d\tau}.$$

It is equivalent in coordinates to the familiar connection equation. It is not a field equation and does not identify the source of the metric. A supported body has nonzero proper acceleration; a

free body can have coordinate acceleration without locally feeling a gravitational force (Einstein, 1916; Geroch and Jang, 1975). Curvature appears operationally through, for example, geodesic deviation,

$$\frac{D^2\xi^\mu}{D\tau^2} = -R^\mu{}_{\nu\rho\sigma}u^\nu\xi^\rho u^\sigma.$$

Universality of free fall, local Lorentz and position behaviour, redshift, light propagation, and post-Newtonian observables constrain any proposed coupling (Will, 2014; Wagner et al., 2012; Touboul et al., 2022; Bertotti, Iess and Tortora, 2003). They do not show that inertial and gravitational concepts or mechanisms are identical.

4 Formal passive-continuation definition

Definition 2.1 is class-relative by construction. A containing frame can price apparatus support or an environmental interaction even where a mechanical-direction class assigns no additional alteration cost to reuse. No sum between classes is licensed without a typed conversion. The definition therefore has no physical units unless a realization supplies them.

Proposition 4.1 (Persistence is not propulsion)**.** *Definition 2.1 supplies no force, physical momentum, time parameter, or equation of motion.*

Proof. The definition fixes a relation in an ordered cost class. It specifies neither a physical state space nor a map to momentum, nor a derivative, symmetry, or interaction law. Each is required to state the listed physical objects. □

The strongest inherited result is thus structural: present operative structure can be reused without an added alteration of that structure in its declared class. Calling this a momentum-like reading is permissible only as a structural analogy. Calling it physical momentum, force, energy, action, mass, or stress–energy requires a separate map with units and measurements.

5 Narrative residual-gradient derivation contract

This section makes the proposed direction of grounding explicit: physical gravity is to be recovered from a realization of present operative Narrative, not used to define the general continuation principle. The following is a new conditional contract. It does not assert that the current corpus already supplies physical gravity, a force in Newtons, or a source equation.

Hypothesis 5.1 (H-IF: residual-gradient realization)**.** *A candidate realization supplies a differential information-state space $\mathcal{Q}_I$, a preparation/readout map*

$$Z : (S_t, \mathcal{H}^W_{[t_0,t]}, \mathcal{R}_t, \mathcal{O}_t) \longrightarrow q \in \mathcal{Q}_I,$$

a physical clock/accelerometer bridge

$$\Theta : (q, \mathcal{O}_t) \longrightarrow (\tau, \mathcal{A}_{\mathrm{acc}}),$$

where τ is an operational physical-clock reading and $\mathcal{A}_{\mathrm{acc}}$ is a declared proper-acceleration/support observable, a residual functional $\mathcal{R}_I : \mathcal{Q}_I \to K_I$, and a declared ordered or differentiable value structure on K_I. Where its derivative is defined, the residual gradient *is the residual derivative*

$$\boxed{\mathcal{F}_I(q) := -D\mathcal{R}_I(q) \in T^*_q\mathcal{Q}_I.} \tag{IF1}$$

A response map $B_q : T^*_q\mathcal{Q}_I \to T_q\mathcal{Q}_I$, if used to select an update direction, must be supplied separately. Thus the residual gradient is intrinsic to the declared information realization: it is the covector of unresolved balance, not a metaphorical name for a later mechanical force.

H-IF has three mandatory physical translation stages. First, it must supply measurement-preserving physical state and clock maps. Second, its Newton and covariant force readouts must be equations, not identifications:

$$\begin{aligned} &K_{\mathrm{N}} : (q, \mathcal{F}_I, \Theta) \longrightarrow \mathbb{R}^3, \qquad [K_{\mathrm{N}}] = \mathrm{kg\,m\,s^{-2}}, \quad K_{\mathrm{N}} = \frac{d\mathbf{p}}{dt}, \\ &K_4 : (q, \mathcal{F}_I, \Theta) \longrightarrow T^*M, \qquad K_4 = \frac{Dp_\mu}{d\tau} \qquad \text{in its declared relativistic convention.} \end{aligned} \tag{IF2}$$

Here Θ supplies the typed clock and accelerometer/readout conditions above. The first line is the required map by which a newton becomes a translation/readout unit for the candidate Narrative force; it does not give a Narrative magnitude SI units by definition. Third, the realization must independently close the existing source and geometry maps

$$\begin{aligned} &L : (q, \mathcal{F}_I, \Theta) \longrightarrow T^{\mathrm{eff}}_{\mu\nu}, \\ &\Phi : (q, \mathcal{F}_I, \Theta) \longrightarrow \\ &\qquad (M, g_{\mu\nu}, \nabla, \mathcal{O}_{\mathrm{phys}}). \end{aligned} \tag{IF3}$$

Each map must state its tensor type and physical dimensions. It must also specify an exchange law, matter coupling, and controlled error. A force/exchange readout is not itself a rank-two gravitational source, and neither source nor geometry may be assumed in order to define $\mathcal{R}_I$.

Proposition 5.2 (Conditional gravity-derivation target)**.** *If one realization satisfies H-IF, closes the maps in (IF2)–(IF3), and proves F1–F6, IG1–IG8, H-PC, history-bearing attribution, universal coupling, and the stated weak-field and precision limits, then that realization derives its claimed physical gravitational regime from the residual-gradient construction. In particular, it must recover the Newtonian force law in its stated limit and the free-path/source/clock relations it claims in the relativistic regime.*

Proof. H-IF supplies a typed residual derivative; (IF2) supplies its testable physical momentum-exchange meaning; and (IF3), together with IG1–IG8, supplies a source, geometry, matter coupling, and observables. F1–F6 establish the flat recovery, while the remaining listed conditions establish the gravitational recovery and its empirical domain. Each arrow is a stated premise, so the conclusion applies only to a realization that proves every map and limit. □

Failure conditions. The contract fails as a derivation if it (i) lacks a declared $\mathcal{Q}_I$, residual, derivative domain, or response/readout rule; (ii) calls $-D\mathcal{R}_I$ a physical newton without proving K_{N}; (iii) inserts $\mathcal{F}_I$ directly into a stress–energy or metric equation without L, Φ, and compatible exchange accounting; (iv) defines the residual using the very gravitational field equation, metric, or force law it claims to derive; or (v) fails the established free/support, tidal, universality, clock, weak-field, or post-Newtonian targets. A candidate observationally identical to GR plus standard low-energy matter remains a grounding interpretation unless it states a finite additional discriminator.

6 Conditional metric-source recovery of general relativity

These recovery gates bound the physics companion's Remark 4.1 and Remark 4.2: calling both masses change-load establishes neither universal coupling nor an equivalence-principle prediction.

This section gives the exact classical GR consequence of a supplied metric-source realization. It does not claim that the generic residual-gradient derivative selects such a realization. In units $c = 1$, let a candidate supply an oriented, time-orientable four-dimensional Lorentzian manifold

(M, g), its Levi–Civita connection, a non-null boundary prescription, matter fields ψ, and a universal metric coupling $S_{\mathrm{m}}[g, \psi]$. For fixed induced boundary metric, the Dirichlet action is

$$\begin{aligned} S_{\mathrm{GR}} &= \frac{1}{2\kappa}\int_M (R - 2\Lambda)\sqrt{|g|}\, d^4x + \frac{1}{\kappa}\int_{\partial M} \varepsilon K \sqrt{|h|}\, d^3x \\ &\quad + S_{\mathrm{joint}} + S_{\mathrm{m}}[g, \psi], \qquad T_{\mu\nu} := -\frac{2}{\sqrt{|g|}}\frac{\delta S_{\mathrm{m}}}{\delta g^{\mu\nu}}, \quad \kappa = 8\pi G. \end{aligned} \tag{GR1}$$

The boundary term and any required joint terms are part of the premise: the bare Einstein–Hilbert integral does not give a Dirichlet variational problem merely by fixing h. $T_{\mu\nu}$ is a symmetric covariant rank-two source obtained from matter metric variation, not a scalar information residual.

For allowed variations, stationarity yields

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}} \qquad \left(= \frac{8\pi G}{c^4} T_{\mu\nu} \text{ when } c \text{ is restored}\right). \tag{GR2}$$

The contracted Bianchi identity is an off-shell geometric identity. Therefore GR2 requires $\nabla^\mu T_{\mu\nu} = 0$ for the total modeled matter source. Diffeomorphism invariance of S_{m} gives the related Noether identity, and on the matter equations supplies this conservation statement; components may exchange energy–momentum and open systems retain boundary flux.

The free-path consequence is separately conditional. With the test-particle premise $S_{\mathrm{pp}} = -m\int ds$, fixed-endpoint worldline variation gives

$$\boxed{u^\nu \nabla_\nu u^\mu = 0.} \tag{GR3}$$

A universal matter coupling is required before interpreting this as a composition-independent free-fall limit. GR2 alone does not make every charged, spinning, extended, self-gravitating, or quantum body exactly geodesic. In a weak, quasi-static, asymptotically Minkowskian regime with $T_{00} \simeq \rho c^2$ and negligible Λ at the scale of interest, GR2 and GR3 recover $\nabla^2\Phi = 4\pi G\rho$ and $\ddot{\mathbf{x}} = -\nabla\Phi$. Clock, light, tidal, and post-Newtonian results require the corresponding metric solution, universal coupling, and stated test domain.

The GE selection breakpoint. The missing map is not GR2's variation but the noncircular selection of GR1:

$$\mathsf{Select}_{\mathrm{GR}} : (\mathcal{F}_I, \mathcal{H}^W, \mathcal{O}, \mathsf{BC}, \mathsf{Units}) \rightrightarrows [g, \nabla, S_{\mathrm{GR}}, S_{\mathrm{m}}, G, \Lambda]_{\mathrm{Diff}(M)}. \tag{GR4}$$

No such map is currently supplied. It must declare its residual codomain, admissible geometry/action class, gauge quotient, boundary/initial conditions, calibration, uncertainty rule, and a uniqueness or empirical selection rule. Writing the Einstein–Hilbert action, a Lorentzian metric, G, Λ, or universal coupling directly into its output constraints assumes the result. Thus GR1–GR3 are an exact conditional GR derivation; GR4 is the remaining information-gravity derivation problem.

6.1 Strongest four-dimensional metric-selection result

The most that an action-selection argument can presently establish is a conditional classification theorem. Suppose a candidate GR4 selector proves, rather than inserts, all of the following restrictions: spacetime is fixed to four dimensions with a declared Lorentzian signature; g is the sole geometrical gravitational field; the bulk metric equation $E_{\mu\nu}[g]$ is a local natural function of the metric two-jet, symmetric, and identically Levi–Civita-divergence-free; no nondynamical background structure occurs; and the Einstein-tensor coefficient is nonzero. Then the four-dimensional Lovelock classification gives

$$\boxed{E_{\mu\nu} = aG_{\mu\nu} + bg_{\mu\nu}.} \tag{GR5}$$

With a separately supplied universal metric matter coupling, division by a puts the field equation into GR2 form with $\Lambda = b/a$ and the relative matter normalization identified as $\kappa = 1/a$ in the chosen conventions. At the action level, the same restricted bulk dynamics is represented by the Einstein–Hilbert term plus a volume term; boundary completion and four-dimensional topological/total-derivative additions remain separately specified.

What GR5 does not select. GR5 does not fix a, b, G, Λ, the metric signature, topology, boundary conditions, matter species, or universal coupling. It also does not justify locality, the metric-only premise, absence of extra fields, or the restriction to second-order metric equations. Relaxing any of those assumptions admits higher-curvature effective-field-theory terms, nonlocal actions, metric-affine/torsional theories, extra fields, or other alternatives. A bare residual-minimization statement cannot supply the restrictions: its domain, norm, measure, admissible fields, and boundary data already carry the selection. In particular, a squared equation residual generally yields higher-order normal equations, while positive reweightings preserve the same zero set. Therefore GR5 narrows the GE selector to a precise proof obligation; it is not a derivation of its premises.

Prior-layer realization obstruction. No generic residual, entropy, or relation principle currently closes the prior realization map from GE to the input of GR5. A physical candidate must first supply: (R1) an event carrier and causal/order relation; (R2) a controlled manifold-like continuum limit with a declared dimension and time orientation; (R3) a volume/scale and observable map that determines a Lorentzian metric rather than a merely conformal or Riemannian structure; and (R4) a dynamical selection rule suppressing nonlocal, nonmetric, higher-derivative, or extra-field alternatives. These are not consequences of an untyped residual. They are the required domain of Φ_ϵ and GR4.

Known conditional routes do not remove this obligation. Horizon-thermodynamic and entanglement-equilibrium arguments begin with Lorentzian causal geometry, area entropy, and a stress tensor; causal-order reconstruction begins with a manifold-like causal order and a volume element; information geometry produces a statistical Riemannian metric; and massless-spin-two consistency begins with a Lorentzian quantum field theory and a graviton. Each can constrain a specified gravitational regime, but none selects Lorentzian four-dimensional, local metric GR and universal matter coupling from a generic information residual alone. This is an underdetermination result, not an impossibility claim about a future specified GE microtheory.

6.2 R1: realized-support proto-causal order

GE can nevertheless construct a narrower combinatorial object. For a declared realization trace, let E^W contain typed realized-transition events

$$e_t = \langle W, G_t, a_t, G_{t+1}, \mathcal{R}_t, \mathcal{O}_t \rangle,$$

where a_t is an actually selected admissible continuation. Define a primitive support link $e \rightsquigarrow e'$ only when the declared output/support of e is a necessary input/support of e' at the stated grain. This is a support relation, not yet a physical influence relation. If the trace supplies (R1a) acyclicity of $\rightsquigarrow$ and (R1b) a finite set of intermediate link-reachable events for each proposed comparable pair at its declared resolution, define

$$e \prec_D e' \quad \Longleftrightarrow \quad e \neq e' \text{ and there is a finite directed } \rightsquigarrow \text{ path from } e \text{ to } e'. \tag{R1}$$

Then $(E^W, \prec_D)$ is a locally finite strict partial order: irreflexivity follows from R1a, transitivity from path concatenation, and local finiteness from R1b. It is invariant under relabeling of events and construction order when all observables depend only on the typed trace and order relation; otherwise construction-order-sensitive observables must remain declared physical or operational data.

Promotion gates. R1 establishes only a candidate *proto-causal* order. To call $\prec_D$ physical causal precedence, a realization must additionally prove that its events are physically individuated rather than states, records, or propositions; that each link denotes propagating possible influence under a declared intervention/dynamical model rather than inference, correlation, or resource support; and that omitted mediators/common causes are controlled. A Lorentzian-order claim further needs a faithful manifoldlike continuum realization with dimension and causal regularity. Under the appropriate continuum hypotheses, causal order fixes only a conformal metric class; a volume/counting calibration, clocks, and observables are separately needed for the full metric. The causal-order uniqueness programme for discrete realizations remains conditional; it cannot be used here as an established geometry theorem.

7 Conditional electromagnetic alternate-grounding

This is a separate conditional programme, not a claim that electromagnetism is already present in the inherited vocabulary. It proposes electromagnetic (EM) physics as a distinct field-edge grounding or physical realization of the same residual-gradient structure in H-IF: it retains the single core residual derivative $\mathcal{F}_I = -D\mathcal{R}_I$ in (IF1), rather than adding a second residual force. The proposed EM realization is neither gravity nor a re-identification of gravitational load-shaped propagation. It has its own state, gauge, matter, source, exchange, boundary, and observable obligations.

The relevant 01a material does not close those obligations. Its $S^1 = U(1)$ discussion motivates a complex amplitude and global phase invariance in a conditional Born-rule recovery; it supplies no local $U(1)$ connection, field strength, charge/current, matter coupling, Maxwell equation, or QED construction. The contingency of a gauge group there is likewise not an EM derivation.

Hypothesis 7.1 (H-EM: electromagnetic field-edge realization)**.** *One candidate supplies, in addition to H-IF, a separately typed electromagnetic state and clock/readout,*

$$X_{\mathrm{EM}} : \mathfrak{C} \longrightarrow \mathcal{X}_{\mathrm{EM}}, \qquad \Theta_{\mathrm{EM}} : \mathcal{X}_{\mathrm{EM}} \longrightarrow (\tau_{\mathrm{EM}}, \mathcal{O}_{\mathrm{EM}}),$$

with preparation, composition, physical units, and relativistic transformation rules. It supplies a local $U(1)$ connection A_μ, its curvature/field strength $F_{\mu\nu}$, and a stated gauge equivalence (for example, $A_\mu \mapsto A_\mu + \partial_\mu \chi$ with the corresponding transformation of charged matter). It separately defines a measurable charge/current (ρ, j^μ), a matter coupling and physical response law, such as the Lorentz-force target $mu^\nu \nabla_\nu u^\mu = qF^\mu{}_\nu u^\nu$, or an explicitly different, testable alternative.

The candidate must give an action and equations, or a viable alternative with equivalent stated conservation and variational/accounting content; gauge- invariant observables (at minimum a declared operational construction from $F_{\mu\nu}$, fluxes, and/or holonomies); interface and boundary conditions; and local and boundary exchange accounting. In a standard covariant realization this includes a current balance such as $\nabla_\mu j^\mu = 0$ and a declared matter–field stress–energy exchange law. It must state relativistic covariance and its domain, and must recover Maxwell theory and QED in their respective tested domains, or name a quantitatively tested alternative with its distinct predictions and error bounds. None of these requirements is supplied by H-IF, H-PC, or 01a.

7.1 Conditional variational recovery of classical electromagnetism

A classical electromagnetic recovery can be derived exactly once H-EM supplies additional geometric and dynamical premises. Let M be an oriented four-dimensional Lorentzian manifold with metric g, let $P \to M$ be a principal $U(1)$ bundle, and let A be a connection with local curvature $F = dA$. Let Φ be charged matter and stipulate the local gauge-invariant field-edge

action

$$S_{\mathrm{edge}}[A,\Phi] = -\frac{1}{4g_{\mathrm{em}}^2}\int_M F_{\mu\nu}F^{\mu\nu}\sqrt{|g|}\,d^4x + S_{\mathrm{m}}[g,A,\Phi], \qquad J^\nu := -\frac{1}{\sqrt{|g|}}\frac{\delta S_{\mathrm{m}}}{\delta A_\nu}. \tag{EM1}$$

These are new realization premises. In particular, neither the choice of $U(1)$, the connection, the quadratic Maxwell term, g_{em}, nor charged matter follows from global phase alone or from the generic residual derivative.

For admissible variations of A, the action gives

$$\begin{aligned}\delta S_{\mathrm{edge}} = &\int_M \sqrt{|g|}\left[g_{\mathrm{em}}^{-2}\nabla_\mu F^{\mu\nu} - J^\nu\right]\delta A_\nu\, d^4x \\ &- g_{\mathrm{em}}^{-2}\int_{\partial M}\sqrt{|h|}\, n_\mu F^{\mu\nu}\delta A_\nu\, d^3x.\end{aligned} \tag{EM2}$$

With a declared Dirichlet boundary condition (or the corresponding added Legendre term and declared flux data), stationarity for arbitrary interior variations therefore yields

$$\boxed{\nabla_\mu(g_{\mathrm{em}}^{-2}F^{\mu\nu}) = J^\nu, \qquad dF = 0.} \tag{EM3}$$

The second equation follows from $F = dA$; the first follows from the stipulated action and source definition. Taking the divergence of the first requires $\nabla_\nu J^\nu = 0$ in the stated smooth classical domain. Equivalently, local gauge invariance plus the matter equations supplies the corresponding current identity, with boundary flux retained rather than silently discarded.

Matter response is another separate premise. Given, for example,

$$S_{\mathrm{pp}} = -m\int ds + q\int A_\mu dx^\mu, \tag{EM4}$$

worldline variation with fixed endpoints yields

$$\boxed{mu^\nu\nabla_\nu u^\mu = qF^\mu{}_\nu u^\nu.} \tag{EM5}$$

Together with (EM3), the electromagnetic stress tensor obeys the matching field–matter exchange relation only when S_{m} is a diffeomorphism-covariant matter action, the matter equations hold, and the specified boundary flux is retained. Thus EM1–EM4 plus those conditions entail the classical Maxwell field equations, charge balance, and the stated Lorentz-force response in their declared domain.

Exact breakpoint. This is a conditional classical-electrodynamics derivation, not a derivation of EM1 from GE. A globally defined phase supplies at most a pure-gauge derivative, not nonzero curvature, a Maxwell kinetic term, a charge representation, or a physical source. The map from $\mathcal{F}_I = -D\mathcal{R}_I$ to the field-edge action must still be supplied without inserting EM1 into the definition of the residual. Quantization, photon states, gauge fixing/BRST or an equivalent physical-state construction, anomaly control, and renormalization are further QED premises; they do not follow from EM3–EM5.

Non-circularity and falsification. H-EM is admissible only if the definition of $\mathcal{R}_I$, the state/clock maps, and the translation from the single $\mathcal{F}_I$ do not assume the Maxwell or QED action, Maxwell equations, a physical A_μ or $F_{\mu\nu}$, charge conservation, the Lorentz-force law, or the experimental EM observables that the realization claims to recover. It fails as a proposed physical realization if it lacks separately typed state/clock, local gauge connection or an explicit viable replacement, charge/current and matter response, an action/equations and gauge-invariant readout, interface/boundary conditions, conservation/exchange accounting, relativistic covariance, or the stated Maxwell/QED-or-alternative recovery. It also fails if a gauge variation changes

an alleged physical observable without a declared symmetry breaking, if the charge/current or field/matter balance is violated in its claimed domain, or if its finite alternative prediction is experimentally excluded. These are falsifiers of H-EM, not of the internal passive-reuse proposition.

This addition does not alter gravity's conditional recovery status. It preserves the limits on local accelerometry and history-bearing attribution, and 01i remains unreleased. It supplies neither physical electromagnetism nor Maxwell theory nor QED; it only makes explicit what a separately implemented EM grounding would have to establish.

8 Flat and curved recovery maps

A candidate realization must present maps rather than metaphors:

$$X : \mathfrak{C} \longrightarrow \mathcal{X}, \qquad P_4 : \mathcal{X} \longrightarrow T^*M,$$
$$\Phi : \mathfrak{C} \longrightarrow (M, g_{\mu\nu}, \nabla, \mathcal{O}_{\text{phys}}), \qquad L : \mathfrak{C} \longrightarrow T^{\text{eff}}_{\mu\nu}.$$

The domains, codomains, units, transformation rules, approximation errors, and operational readouts of all four maps are obligations. The project currently supplies none of them. In particular, a graph is not a Lorentzian manifold, an alteration cost is not stress–energy, and cycle order is not proper time.

8.1 Flat-map target

The following named conditions are required to label the flat equation a recovery rather than an assumption.

Condition 8.1 (F1–F6: flat passive-transport conditions)**.** *A realization must provide: (F1) a flat spacetime structure $(M, \eta_{\mu\nu}, \nabla^{(\eta)})$ with declared signature, physical state space, and preparation/readout protocol; (F2) a dimensioned covector momentum map $P_4 : \mathcal{X} \to T^*M$, mass-shell convention, and the raised vector $p^\mu = \eta^{\mu\nu} p_\nu$ with its Lorentz transformation law; (F3) a proper-time or declared affine-parameter clock map and its timelike/massless domain; (F4) a free matter action or equations of motion; (F5) the translation symmetry or explicitly stated balance structure that supports the claimed conservation; and (F6) an isolation and boundary-flux domain. Only if F1–F6 yield*

$$\boxed{\frac{dp^\mu}{d\tau} = 0}$$

for the declared free flat regime is this a flat passive-continuation recovery target met. The equation is not derived in this paper.

The condition does not confer global momentum conservation on an arbitrary curved spacetime or on a subsystem. It requires a current, symmetry, or stated replacement and explicit field/boundary exchange.

8.2 Load-shaped-map target

Condition 8.2 (IG1–IG8: load-shaped free-propagation conditions)**.** *A realization must provide: (IG1) a controlled map to a Lorentzian metric $g_{\mu\nu}$ and a declared connection ∇, including whether it is Levi–Civita or the constraint that determines otherwise; (IG2) a causal and proper-time clock observable; (IG3) a matter coupling and test-body approximation; (IG4) a derivation of the metric geodesic law from that coupling; (IG5) a dimensioned operational load/source map to $T^{\text{eff}}_{\mu\nu}$; (IG6) a metric field equation or explicitly parameterized metric modification; (IG7) compatible*

source/exchange and boundary accounting; and (IG8) tidal and accelerometer observables. Only if IG1–IG8 yield

$$\boxed{u^\nu \nabla_\nu u^\mu = 0}$$

in the stated free-body regime is this a load-shaped geodesic recovery target met. The equation is not derived in this paper.

A geodesic target alone supplies neither the metric's source nor its dynamics. For an Einstein-tensor form, the contracted Bianchi identity requires a compatible source/exchange relation. A candidate must give the tensor type, units, operational definition, and any $\nabla^\mu T^{\text{eff}}_{\mu\nu} = J_\nu$ accounting; it cannot insert an untyped cost into a field equation (Einstein, 1916; Brading, 2005; Szabados, 2009).

9 Equivalence and source obligations

Hypothesis 9.1 (H-PC: physical passive-continuation bridge)**.** *One concrete realization assigns the same declared physical passive-reuse sector to both the F1–F6 flat regime and the IG1–IG8 load-shaped regime, with all maps in Section 8 defined and measured.*

H-PC is a new hypothesis, not an inherited result. It owes a universal matter coupling calculation before it can invoke the equivalence principle. That calculation must state inertial and gravitational response, composition, binding-energy, spin/history dependence, clock coupling, and the predicted Eotvos parameter or corresponding observable. Existing tests tightly constrain such departures (Will, 2014; Wagner et al., 2012; Touboul et al., 2022).

The source obligation is separate. The model must identify a dimensioned source $T^{\text{eff}}_{\mu\nu}$, its exchange law, and metric gravity dynamics. It must recover or explicitly parameterize deviations from the weak-field targets

$$\nabla^2 \Phi_N = 4\pi G\rho, \qquad \ddot{\mathbf{x}} = -\nabla \Phi_N$$

and the applicable redshift, light, clock, and post-Newtonian limits. Neither universal coupling nor geodesic motion derives Einstein equations; neither a source equation nor a metaphorical "load" derives a geodesic.

10 Four-way derivation ledger

Inherited/internal results	New hypotheses	Open physical bridge obligations	Empirical constraints
Passive reuse has zero additional alteration cost in its declared class; it is not zero total activity or time.	H-PC: one physical realization has common passive-reuse sectors in both regimes.	Give state identity, physical alteration class, units, support, and operational readout.	Candidate must state which activity, clock, and support measurements distinguish class cost from total expenditure.
Momentum is not force; persistence alone does not prove momentum conservation.	H-F: carried structure maps to measured four-momentum in a flat regime.	Close F1–F6 and derive $dp^\mu/d\tau = 0$, including symmetry, isolation, and flux accounting.	Recover inertial transport and measured momentum balance within the declared special-relativistic/ Newtonian domain.
Free GR test-body motion has zero proper acceleration; support and tides remain real.	H-G: a dimensioned load maps to controlled geometry and matter coupling.	Close IG1–IG8 and derive $u^\nu \nabla_\nu u^\mu = 0$ plus source dynamics.	Meet accelerometer, tidal, weak-field, redshift, light-propagation, and PPN limits.
A present passive continuation is historically underdetermined; proper acceleration is only a local relation.	H-I: a history-bearing attribution record has a specified inference relation.	Define retained changes, source/support history, uncertainty, scale, and comparison protocol without a privileged observer.	Match local accelerometer results while testing momentum/load attribution through the declared historical record.
No project source, metric, connection, equivalence principle, or conservation theorem has been established.	H-U: the coupling is universal in the stated domain.	Calculate composition, binding-energy, spin, and clock effects; give Bianchi-compatible exchange.	Satisfy UFF/EEP and local-Lorentz constraints or state a finite excluded parameter range.

11 Prior art

Newton provides the historical inertial/change-law setting; Noether gives the symmetry–conservation theorem under its action hypotheses (Newton, 1687; Noether, 1918). Einstein supplies the metric field theory, while Geroch and Jang formalise a geodesic principle for suitably small bodies under stated hypotheses (Einstein, 1916; Geroch and Jang, 1975). These sources give targets in their own formalisms; none maps this project's alteration cost or carried history to physical momentum or geometry.

Machian and relational-inertia proposals motivate a comparison but do not close this bridge (Sciama, 1953; Barbour and Bertotti, 1977). Teleparallel gravity gives a reformulation of already specified GR variables; it is not a license to call gravity ordinary flat-space inertia, nor a

derivation from passive continuation (Aldrovandi and Pereira, 2013; Li, Sotiriou and Barrow, 2011). Effective field theory likewise constrains low-energy departures after fields and couplings have been defined; it does not supply them (Donoghue, 1994; Burgess, 2004).

12 Recovery conditions and falsifiers

The conditional reconciliation is met only if one realization closes F1–F6 and IG1–IG8, supplies H-PC's common implemented dynamics, calculates universal coupling and source exchange, and recovers the declared tested limits. Even then it establishes two controlled readings of a law, not a force identity.

A candidate fails as a physical reconciliation if any of the following holds:

1. its alteration class lacks declared comparison, units or ordering, support, and a measurement protocol;
2. its free flat regime fails to yield constant measured momentum/four-momentum under its stated isolation conditions, or it asserts conservation without a symmetry/current and boundary accounting;
3. its load has no dimensioned source type, or its geometry/source equation has incompatible divergence or unaccounted exchange;
4. its clock, proper-acceleration, free/support, or tidal observable conflicts with its claimed regime's established tests;
5. its matter coupling violates UFF, local-Lorentz, redshift, or post-Newtonian constraints without a surviving quantitative fit;
6. it uses a metric, geodesic principle, universal coupling, conservation law, or stress tensor as an input while calling it a passive-continuation derivation; or
7. it attributes a present passive continuation to momentum or gravity without the declared history-bearing record and inference map; or
8. it has no finite discriminator beyond GR plus standard low-energy matter while claiming a new physical theory. In that case it is, at most, an interpretation subject to its own coherence audit.

13 Conclusion

This paper establishes one internal result and one strict programme condition. The result is Definition 2.1: reuse of presently operative structure has zero additional alteration cost in a declared grounding class. The condition is H-PC: flat inertial transport and load-shaped geodesic motion count as two physical readings of a common passive-continuation law only after a concrete realization derives their equations and limits through the named maps and conditions.

Nothing in the inherited vocabulary derives physical momentum, its conservation, a metric, connection, source, equivalence principle, geodesic equation, or quantum-gravity result. Momentum remains distinct from force. GR free fall remains distinct from a locally felt gravitational force, while support and tidal effects remain observable. A present in-frame record can establish only its declared local relation; differentiating momentum from gravity as an explanatory attribution requires a history-bearing record and the stated inference map. That requirement supplies no contrary identity claim.

H-EM adds a distinct, conditional electromagnetic field-edge programme while preserving those limits. It retains one residual-gradient residual derivative and requires separately typed

state/clock, local gauge, charge/current, matter, dynamics, observables, interface, exchange, covariance, and empirical-recovery bridges. It does not derive physical electromagnetism, Maxwell theory, or QED; 01i remains unreleased, and gravity recovery remains conditional.

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